

A Parabolic Gap Theorem on the Four-Sphere

1 Introduction

This note serves as a summary of the paper “Parabolic Gap Theorems for the Yang–Mills Energy” by Anuk Dayaprema and Alex Waldron. Details and results omitted here can be found directly in their original article.

In this note we explain a parabolic gap theorem for Yang–Mills flow on the round four-sphere. The main result states that if the initial Yang–Mills energy lies below the first non-instanton Yang–Mills level, then the flow exists for all time and converges exponentially to an instanton.

This result may be viewed as a dynamical analogue of the classical elliptic gap theorem. In the elliptic setting, one starts with a Yang–Mills connection and proves that below a certain threshold it must already be anti-self-dual. Here the initial connection is arbitrary, and the conclusion is instead about the long-time behavior of the flow

$$\partial_t A = -D_A^* F_A. \quad (1)$$

The question is whether a low-energy initial connection must evolve toward an instanton.

On an oriented four-manifold, the curvature splits as

$$F_A = F_A^+ + F_A^-,$$

and instantons are characterized by the condition $F_A^+ = 0$. The low-energy assumption therefore gives control of the self-dual curvature in L^2 . The main difficulty is that this is not yet enough to rule out singularity formation, since blow-up is a pointwise phenomenon.

The key step in the proof is a local split ε -regularity theorem, which converts a low L^2 bound for F_A^+ into pointwise control. Once this local estimate is established, one obtains global control of the flow and hence long-time existence. One then extracts a Yang–Mills limit and uses the gap condition to show that this limit must be anti-self-dual. Finally, the special geometry of the round sphere yields exponential convergence.

Thus the proof follows the chain

$$\text{energy gap} \implies \text{pointwise control of } F_A^+ \implies \text{global existence} \implies \text{Yang–Mills limit} \implies \text{instanton}.$$

2 Energy, instantons, and the gap level

Let G be a compact Lie group, and let $\rho : G \rightarrow \mathrm{GL}_{\mathbb{K}}(V)$ be a finite-dimensional orthogonal or unitary representation, where $\mathbb{K} = \mathbb{R}$ or $\mathbb{C}$. Let $P \rightarrow S^4$ be a principal G -bundle, and let $E = P \times_{\rho} V$ be the associated vector bundle. A connection on P induces a covariant derivative on E , and by abuse of notation we denote both by the same letter. Accordingly, when we speak of a connection A below, we may regard it either as a principal connection on P or as the induced connection on E . Let $\mathfrak{g}$ be the Lie algebra of G . The adjoint bundle of P is $\mathfrak{g}_E := P \times_{\mathrm{Ad}} \mathfrak{g}$, and via the differential of the representation ρ , we view $\mathfrak{g}_E$ as a subbundle of $\mathrm{End}(E)$. Hence the curvature of A is a $\mathfrak{g}_E$ -valued 2-form,

$$F_A \in \Omega^2(S^4, \mathfrak{g}_E) \subset \Omega^2(S^4, \mathrm{End}(E)).$$

Since the base is four-dimensional, the Hodge star operator gives the decomposition

$$F_A = F_A^+ + F_A^-,$$

where F_A^+ and F_A^- are respectively the self-dual and anti-self-dual parts of the curvature. With the chosen orientation, an instanton means an anti-self-dual connection, namely

$$F_A^+ = 0.$$

The Yang–Mills energy is

$$\text{YM}(A) = \frac{1}{2} \int_{S^4} |F_A|^2 d\text{vol}.$$

A Yang–Mills connection is a critical point of the Yang–Mills functional, or equivalently a solution of

$$D_A^* F_A = 0.$$

To relate this to topology, one introduces the characteristic number $\kappa(E)$. Following Donaldson–Kronheimer, we define

$$\kappa(E) = \begin{cases} \langle c_2(E) - \frac{1}{2}c_1(E)^2, [S^4] \rangle, & \mathbb{K} = \mathbb{C}, \\ -\frac{1}{4}\langle p_1(E), [S^4] \rangle, & \mathbb{K} = \mathbb{R}. \end{cases}$$

By Chern–Weil theory, for any connection A on E one has

$$\kappa(E) = \frac{1}{8\pi^2 n_{\mathbb{K}}} \int_{S^4} \text{Tr}_{\mathbb{K}}(F_A \wedge F_A),$$

where

$$n_{\mathbb{K}} = \begin{cases} 1, & \mathbb{K} = \mathbb{C}, \\ 4, & \mathbb{K} = \mathbb{R}. \end{cases}$$

From the decomposition of F_A into self-dual and anti-self-dual parts, one obtains the Chern–Weil energy identities

$$\text{YM}(A) = 4\pi^2 n_{\mathbb{K}} \kappa(E) + \|F_A^+\|_{L^2(S^4)}^2 = -4\pi^2 n_{\mathbb{K}} \kappa(E) + \|F_A^-\|_{L^2(S^4)}^2. \quad (2)$$

In particular,

$$\text{YM}(A) \geq 4\pi^2 n_{\mathbb{K}} |\kappa(E)|. \quad (3)$$

Thus $4\pi^2 n_{\mathbb{K}} |\kappa(E)|$ is the topological minimum allowed by Chern–Weil theory. With the orientation fixed so that instantons are characterized by $F_A^+ = 0$, the quantity $\|F_A^+\|_{L^2}^2$ is exactly the part of the Yang–Mills energy beyond the topologically minimal value.

To formulate a uniform energy threshold for non-instanton Yang–Mills connections on S^4 , one introduces a constant depending only on the structure group data (G, ρ) . This quantity is defined by taking the infimum of the self-dual energy among all non-instanton Yang–Mills connections arising from principal G -bundles over S^4 .

Definition 2.1. Let $\mathcal{B}_{S^4, G}$ denote the space of gauge-equivalence classes of smooth connections carried by principal G -bundles $P \rightarrow S^4$. For $[B] \in \mathcal{B}_{S^4, G}$, let B_ρ denote the induced connection on the associated bundle $E = P \times_\rho V$, we define

$$\delta_{G, \rho} = \inf \left\{ \frac{1}{4\pi^2 n_{\mathbb{K}}} \|F_{B_\rho}^+\|_{L^2(S^4)}^2 : [B] \in \mathcal{B}_{S^4, G}, D_B^* F_B = 0, F_{B_\rho}^+ \neq 0 \right\}. \quad (4)$$

Thus $\delta_{G, \rho}$ is the first positive Yang–Mills gap above the instanton level for the structure group (G, ρ) .

By definition, $\delta_{G, \rho}$ is the largest number for which the following implication holds: if B is a Yang–Mills connection on E and

$$\|F_B^+\|_{L^2(S^4)}^2 < 4\pi^2 n_{\mathbb{K}} \delta_{G, \rho},$$

then necessarily

$$F_B^+ = 0.$$

In other words, below this threshold there are no non-instanton Yang–Mills connections.

The global theorem can now be stated as follows.

Theorem 2.2. *Let $k \geq 2$, let $E \rightarrow S^4$ be a (G, ρ) -bundle defined above, and let A_0 be an H^k connection on E satisfying*

$$\text{YM}(A_0) < 4\pi^2 n_{\mathbb{K}}(|\kappa(E)| + \delta_{G,\rho}). \quad (5)$$

Then the Yang–Mills flow with initial value A_0 exists for all time and converges exponentially in H^k to an instanton on E .

This theorem has a clear interpretation. The term $4\pi^2 n_{\mathbb{K}}|\kappa(E)|$ is the topological instanton level coming from Chern–Weil theory, while $\delta_{G,\rho}$ measures the first possible positive gap above that level among Yang–Mills critical points. Hence the assumption says that the initial energy lies strictly below the first non-instanton Yang–Mills level.

The proof does not begin with the global theorem itself. The above energy bound only gives an L^2 control on the self-dual curvature, while continuation of the flow depends on pointwise curvature bounds. Thus one first proves a local theorem which prevents concentration of F^+ under the gap threshold.

3 The local split ε -regularity theorem

The local theorem is the analytic starting point of the global argument. Its role is to show that, under the gap assumption, a uniform-in-time L^2 bound for the self-dual curvature cannot concentrate locally, and hence upgrades to pointwise control.

Theorem 3.1. *Fix $\gamma > 0$. Then there exist $k \in \mathbb{N}$ and a constant $C_{(3.1)} > 0$, depending only on γ and (G, ρ) , with the following property.*

Let $B = B_1(0)$ be a geodesic ball for a smooth metric g which is sufficiently close to the Euclidean metric in C^k . Suppose $A(t)$ is a smooth Yang–Mills flow on a (G, ρ) -bundle $E \rightarrow B$ for $t \in (-1, 0)$, and assume

$$\sup_{-1 < t < 0} \text{YM}(A(t)) \leq \gamma^{-1} \quad (6)$$

and

$$\sup_{-1 < t < 0} \|F_A^+(t)\|_{L^2(B)}^2 \leq \delta \leq 4\pi^2 n_{\mathbb{K}} \delta_{G,\rho} - \gamma, \quad (7)$$

where self-duality and norms are defined with respect to g . Then

$$\sup_{B_{1/2} \times [-1/4, 0]} |F_A^+| \leq C_{(3.1)} \sqrt{\delta}. \quad (8)$$

The proof uses two auxiliary inputs. The first says that if the self-dual curvature has very small L^2 -mass on a ball at one earlier time-slice, then it cannot become very large immediately afterwards. We will use this only through its contrapositive: a large later pointwise value forces a definite earlier amount of L^2 -energy.

Lemma 3.2. *There exists a constant $0 < c < \frac{1}{10}$, depending only on the structure group (G, ρ) , with the following property. Let $A(t)$ satisfy the assumptions of Theorem 3.1. If $0 < r \leq 1$ and*

$$\int_{B_r} |F_A^+(-(cr)^2)|^2 dV \leq c^2, \quad (9)$$

then

$$\sup_{B_{r/2} \times [-\frac{1}{2}(cr)^2, 0]} |F_A^+| \leq \frac{1}{cr^2}. \quad (10)$$

The second input is a standard point-selection lemma. Its purpose is to replace an arbitrary large-curvature point by a nearby point whose curvature is still large but is also uniformly controlled on its own natural scale. That normalization is what makes compactness available.

For $r > 0$, denote

$$P_r(x, t) := B_r(x) \times [t - r^2, t].$$

Lemma 3.3. *Let $\lambda > 0$, and let f be a non-negative continuous function on $P_r(x_0, t_0)$ such that $f(x_0, t_0) = \lambda^{-2} > 16r^{-2}$. Then there exists a spacetime point $(\tilde{x}, \tilde{t}) \in P_{2\lambda}(x_0, t_0)$ such that, writing $\mu^{-2} := f(\tilde{x}, \tilde{t})$, one has $\mu \leq \lambda$ and*

$$\sup_{P_\mu(\tilde{x}, \tilde{t})} f \leq 4\mu^{-2}. \quad (11)$$

Proof of Theorem 3.1. The argument has two parts. We first prove a rough pointwise bound

$$\sup_{B_{1/2} \times [-1/4, 0)} |F_A^+| \leq C, \quad (12)$$

with C depending only on γ and (G, ρ) , but not on δ . The reason for starting with this weaker estimate is that a direct iteration from the L^2 hypothesis is not robust enough to rule out concentration. Instead, if (12) failed, one could blow up around points of large self-dual curvature and try to extract a nontrivial Yang–Mills limit. The point of the gap condition is precisely that such a limit cannot exist. Once concentration has been excluded in this way, the equation for F_A^+ becomes amenable to a standard parabolic Moser iteration, which upgrades the rough bound to the sharp estimate with the factor $\sqrt{\delta}$.

Step 1: contradiction sequence for the rough bound. Suppose that (12) is false. Then there exists a sequence of smooth Yang–Mills flows $A_i(t)$, with metrics g_i converging to the Euclidean metric in C^∞ , such that

$$\sup_{B_{1/2} \times [-1/4, 0)} |F_{A_i}^+| \geq c^{-4i-1},$$

where c is the constant from Lemma 3.2. After recentering the spacetime maximum and parabolically rescaling by a factor no less than $\frac{1}{2}$, which preserves the form of the assumptions, we may assume

$$|F_i^+(0, 0)| \geq c^{-4i-1},$$

where $F_i := F_{A_i}$.

The motivation for this normalization is simple: we want the large-curvature point to sit at a fixed spacetime location, so that later rescalings can be compared on a common domain.

Step 2: find a time interval on which the flow is almost stationary. The blow-up argument needs not only large curvature, but also an approximately time-independent profile after rescaling. This is obtained from the local energy inequality for the self-dual curvature. More precisely, if $\varphi \in C_c^\infty(B_1)$ is a cutoff function and $-1 \leq a \leq b < 0$, then the Yang–Mills flow equation

$$\partial_t A_i = -D_i^* F_i$$

implies, after testing the evolution of F_i^+ against $\varphi^2 F_i^+$ and integrating by parts, an inequality of the form

$$\int_B \varphi^2 |F_i^+(b)|^2 + \frac{1}{2} \int_a^b \int_B \varphi^2 |D_i^* F_i|^2 \leq \int_B \varphi^2 |F_i^+(a)|^2 + C \int_a^b \int_B |\nabla \varphi|^2 |F_i^+|^2. \quad (13)$$

Choosing $\varphi \equiv 1$ on $B_1(0)$ and supported slightly inside the ambient ball, and then using the uniform bound (7), we obtain a spacetime estimate of the form

$$\int_{P_1(0,0)} |D_i^* F_i|^2 dV dt \leq C\delta. \quad (14)$$

This controls the total dissipation on the whole unit cylinder. The next task is to locate one shorter time interval on which the dissipation is much smaller than average.

Define

$$t_j := -c^{4j}, \quad I_j := [t_{j-1}, t_j]. \quad (15)$$

These intervals are essentially disjoint, and for each i there are about $i/2$ of them in the range

$$j \in \left\{ \left\lfloor \frac{i}{2} \right\rfloor, \dots, i \right\}.$$

Hence, by the pigeonhole principle, there exists $j_i \in \{\lfloor i/2 \rfloor, \dots, i\}$ such that

$$\int_{B_1(0) \times I_{j_i}} |D_i^* F_i|^2 dV dt \leq \frac{2C\delta}{i}. \quad (16)$$

This is the first decisive choice in the proof: it identifies a scale on which the flow is almost stationary in an averaged sense.

Let

$$\tau_i := t_{j_i}.$$

Step 3: convert the large final value into definite earlier L^2 -mass. Why do we go back to the earlier time τ_i ? Because Lemma 3.2 compares an L^2 -norm at time $-(cr)^2$ with a later pointwise bound. Once τ_i is chosen, the matching spatial scale is forced by

$$(cr)^2 = -\tau_i, \quad \text{so} \quad r = c^{-1} \sqrt{-\tau_i}.$$

If one had

$$\int_{B_{c^{-1}\sqrt{-\tau_i}}(0)} |F_i^+(\tau_i)|^2 dV \leq c^2,$$

then Lemma 3.2 would imply

$$|F_i^+| \leq \frac{1}{cr^2} = \frac{1}{-\tau_i} = c^{-4j_i}$$

on a later smaller parabolic cylinder, contradicting

$$|F_i^+(0, 0)| \geq c^{-4i-1},$$

Therefore

$$\int_{B_{c^{-1}\sqrt{-\tau_i}}(0)} |F_i^+(\tau_i)|^2 dV > c^2. \quad (17)$$

This is exactly where the first auxiliary lemma is used: a large value of $|F_i^+|$ at the final time implies that, at the corresponding earlier time and spatial scale, the L^2 norm of F_i^+ cannot be too small.

By averaging (17), there exists a point $\tilde{x}_i \in B_{c^{-1}\sqrt{-\tau_i}}(0)$ such that

$$\lambda_i^{-2} := |F_i^+(\tilde{x}_i, \tau_i)| > \frac{c^3}{C(-\tau_i)}. \quad (18)$$

Step 4: point selection and intrinsic normalization. A point where curvature is large is still not enough for compactness: the curvature could be much larger at nearby points, so rescaling there might lose control immediately. The role of point selection is to move to a nearby point whose curvature is large and uniformly controlled on its own natural scale.

Set $r_i^2 := (c^{-4} - 1)(-\tau_i)$, using (18), one checks that, for $c > 0$ chosen sufficiently small depending only on (G, ρ) ,

$$r_i^2 \lambda_i^{-2} > 16.$$

Hence Lemma 3.3 applies to $f = |F_i^+|$ on $P_{r_i}(\tilde{x}_i, \tau_i)$. We obtain a spacetime point $(x_i, s_i) \in P_{2\lambda_i}(\tilde{x}_i, \tau_i)$ such that, with $\mu_i^{-2} := |F_i^+(x_i, s_i)|$, one has

$$\sup_{P_{\mu_i}(x_i, s_i)} |F_i^+| \leq 4\mu_i^{-2}. \quad (19)$$

By construction, $\mu_i \leq \lambda_i \leq \frac{r_i}{4}$, and from (18) together with $j_i \geq \lfloor i/2 \rfloor$ we get $\lambda_i \rightarrow 0$. Moreover,

$$d_{g_i}(x_i, 0) \leq d_{g_i}(x_i, \tilde{x}_i) + d_{g_i}(\tilde{x}_i, 0) \leq 2\lambda_i + c^{-1}\sqrt{-\tau_i} \rightarrow 0.$$

Hence, for i sufficiently large,

$$P_{\mu_i}(x_i, s_i) \subset B_{1/2}(x_i) \times [s_i - \mu_i^2, s_i] \subset B_1(0) \times I_{j_i}. \quad (20)$$

This containment is important for two reasons: it keeps the rescaling inside the original domain, and it ensures that the small dissipation estimate (16) will survive the blow-up.

Step 5: blow up and extract a Yang–Mills limit. We now parabolically rescale about (x_i, s_i) by the factor μ_i . Denote the rescaled flows by $\tilde{A}_i$. Then on $P_1(0, 0)$ we have

$$|\tilde{F}_i^+(0, 0)| = 1$$

and, by (19),

$$\sup_{P_1(0,0)} |\tilde{F}_i^+| \leq 4.$$

This is exactly the normalization we wanted: the curvature is nontrivial at the origin, but uniformly controlled on a fixed unit parabolic cylinder.

Because the spacetime integral of $|D^*F|^2$ is scale-invariant in dimension four, the small dissipation on $B_1(0) \times I_{j_i}$ transfers to the rescaled cylinder. Using (20) and (16), we get

$$\int_{P_1(0,0)} |D_{\tilde{A}_i}^* \tilde{F}_{\tilde{A}_i}|^2 dV dt \rightarrow 0. \quad (21)$$

So after rescaling, the flow becomes asymptotically stationary.

Standard Uhlenbeck compactness now applies. After passing to a subsequence and choosing local gauges, the rescaled connections converge smoothly on compact subsets of $(\mathbb{R}^4 \setminus \{y_1, \dots, y_\ell\}) \times (-1, 0)$ to a limit connection A_∞ . The vanishing (21) implies that A_∞ is Yang–Mills:

$$D_{A_\infty}^* F_{A_\infty} = 0.$$

Moreover, the limit is nontrivial at the base point. Since the origin is not a singular point of the limiting process, the smooth convergence gives

$$|F_{A_\infty}^+(0, 0)| = \lim_{i \rightarrow \infty} |\tilde{F}_i^+(0, 0)| = 1.$$

By continuity of $F_{A_\infty}^+$, there exist $r_0 > 0$ and $c_0 > 0$ such that

$$\int_{B_{r_0}(0)} |F_{A_\infty}^+(x, 0)|^2 dV \geq c_0. \quad (22)$$

In particular, $F_{A_\infty}^+ \not\equiv 0$, so A_∞ is not anti-self-dual.

By removable singularity at the finite points and at infinity, the limit extends to a smooth Yang–Mills connection, still denoted A_∞ , on a possibly different (G, ρ) -bundle over S^4 .

Step 6: contradiction with the gap level. This is where the global input enters. The self-dual L^2 -energy is conformally invariant in dimension four, so the bound on $\|F_A^+\|_{L^2}$ is preserved under the blow-up. Using Fatou’s lemma and the uniform hypothesis on the original sequence, we obtain

$$\|F_{A_\infty}^+\|_{L^2(S^4)}^2 \leq 4\pi^2 n_{\mathbb{K}} \delta_{G,\rho} - \gamma. \quad (23)$$

By the defining property of $\delta_{G,\rho}$, any Yang–Mills connection on S^4 whose self-dual energy lies strictly below $4\pi^2 n_{\mathbb{K}} \delta_{G,\rho}$ must be anti-self-dual. Hence $F_{A_\infty}^+ \equiv 0$, which contradicts (22).

This contradiction proves the rough bound (12).

Step 7: from the rough bound to the sharp $\sqrt{\delta}$ estimate. Once the blow-up phenomenon has been ruled out, the remaining argument is standard. The rough bound gives

$$\sup_{P_{3/4}(0,0)} |F_A^+| \leq C,$$

so the self-dual curvature satisfies a linear parabolic inequality with bounded coefficients,

$$(\partial_t - \Delta)|F_A^+| \leq C|F_A^+|.$$

The reason this now helps is that the coefficients are controlled independently of δ , while the assumed L^2 -norm of F_A^+ is of size δ . A parabolic Moser iteration therefore upgrades the time-uniform L^2 -control to the desired L^∞ -control:

$$\sup_{B_{1/2} \times [-1/4, 0]} |F_A^+| \leq C_{(3.1)} \sqrt{\delta}. \quad (24)$$

This proves the theorem. $\square$

4 Global existence and convergence to an instanton

We now explain how the local theorem is used to derive the global flow theorem. The overall strategy is as follows. The gap assumption gives a uniform L^2 -bound for the self-dual curvature F_A^+ , but this by itself is too weak for global continuation, since finite-time singularities are detected by pointwise blow-up of the full curvature. Thus the first task is to convert the L^2 -control of F_A^+ into a global pointwise bound. Once this is done, a continuation estimate shows that the full curvature cannot blow up in finite time, and hence the flow exists for all time. Only after long-time existence has been established do we turn to the asymptotic behavior as $t \rightarrow \infty$.

Let $A(t)$ be the Yang–Mills flow starting from A_0 . Since the Yang–Mills energy is nonincreasing along the flow, the initial gap condition is preserved for all positive time:

$$\text{YM}(A(t)) < 4\pi^2 n_{\mathbb{K}} (|\kappa(E)| + \delta_{G,\rho}).$$

By the Chern–Weil identity, this is equivalent to

$$\|F_A^+(t)\|_{L^2(S^4)}^2 < 4\pi^2 n_{\mathbb{K}} \delta_{G,\rho}$$

for all $t > 0$. So the flow stays uniformly below the Yang–Mills gap threshold at every positive time.

The significance of this observation is that it places the flow exactly in the range where the local split ε -regularity theorem applies. What that theorem gives is not yet a global statement, but a local pointwise estimate on any sufficiently small parabolic cylinder. To obtain a global consequence, one applies it around an arbitrary spacetime point after rescaling to a fixed unit cylinder.

Indeed, since the round metric is smooth, there exists a universal radius $R_0 > 0$ such that every geodesic ball $B_{R_0}(x) \subset S^4$, after rescaling to unit size, has metric sufficiently close to the Euclidean metric for Theorem 3.1 to apply. Now fix $x \in S^4$ and $t > 0$, and set

$$r := \min\{R_0, \sqrt{t}\}.$$

This choice has two purposes. The bound $r \leq R_0$ ensures that, after rescaling, the metric is close enough to the Euclidean metric, while the bound $r \leq \sqrt{t}$ guarantees that the parabolic cylinder

$$B_r(x) \times [t - r^2, t]$$

lies inside the time interval where the flow is defined. After rescaling this cylinder to unit size, the local theorem applies to the rescaled flow. Since the L^2 -norm of F_A^+ is scale-invariant in dimension four, the smallness hypothesis is preserved under the rescaling, and therefore the rescaled self-dual curvature is uniformly bounded at the base point. Scaling back to the original variables gives

$$|F_A^+(x, t)| \leq Cr^{-2}.$$

Since $r = \min\{R_0, \sqrt{t}\}$, we obtain

$$|F_A^+(x, t)| \leq \frac{C}{\min\{1, t\}} \quad \text{for all } x \in S^4, t > 0. \quad (25)$$

This is the first global consequence of the local theorem: a purely local estimate has been turned into a global pointwise control of the self-dual curvature.

The next goal is to rule out finite-time singularities. For this, controlling only F_A^+ is not enough, because continuation of Yang–Mills flow depends on the full curvature F_A . Thus one needs a mechanism which propagates control of F_A^+ into control of F_A itself. This is provided by the continuation estimate: if

$$K = \int_{t_1}^{t_2} \|F^+(t)\|_{L^\infty} dt$$

is finite, then

$$\|F(t_2)\|_{L^\infty} \leq \max \left\{ R_0^{-2}, e^{C \max\{1, K\}} \|F(t_1)\|_{L^\infty} \right\}. \quad (26)$$

The point of introducing K is that the estimate (25) makes $\|F^+(t)\|_{L^\infty}$ integrable on every finite time interval. Hence the right-hand side of (26) stays finite on every bounded interval, so the full curvature cannot blow up in finite time. Therefore the Yang–Mills flow exists for all time.

Once long-time existence is known, the problem changes in nature. At this stage one no longer asks whether the flow survives, but what it converges to. The natural first step is to extract a subsequential limit. Choose any sequence $t_i \rightarrow \infty$. By Uhlenbeck compactness and removable singularity, a subsequence converges to a Yang–Mills connection A'_∞ on S^4 . It is worth noting that this limiting connection does not necessarily exist on the original bundle. Therefore, if we ultimately intend for this connection to be the instanton attained by the Yang–Mills flow, a rigorous proof is still required.

Now the gap condition becomes decisive once again. Since the L^2 -norm of F_A^+ is nonincreasing along the flow, it passes to the limit and gives

$$\|F_{A'_\infty}^+\|_{L^2(S^4)}^2 \leq \|F_{A_0}^+\|_{L^2(S^4)}^2 < 4\pi^2 n_{\mathbb{K}} \delta_{G, \rho}. \quad (27)$$

But A'_∞ is itself Yang–Mills, so the definition of $\delta_{G, \rho}$ forces

$$F_{A'_\infty}^+ = 0.$$

Thus every Yang–Mills limit of the flow is an instanton. This is the key structural conclusion: the flow cannot converge, even along a subsequence, to any non-instanton Yang–Mills critical point.

At this point we know that some sequence of times satisfies

$$\|F_A^+(t_i)\|_{L^2(S^4)} \rightarrow 0.$$

The role of monotonicity is now to upgrade this subsequential information to a statement about the whole flow. Since $\|F_A^+(t)\|_{L^2(S^4)}^2$ is nonincreasing in t , once it becomes small along one sequence, it must in fact tend to zero for all times:

$$\|F_A^+(t)\|_{L^2(S^4)} \rightarrow 0 \quad \text{as } t \rightarrow \infty. \quad (28)$$

So the entire flow becomes asymptotically anti-self-dual, not merely a subsequence.

This still gives only qualitative convergence. To obtain a quantitative decay rate, one uses the Weitzenböck formula for self-dual 2-forms on the round sphere. The reason this works especially well on S^4 is that the round metric has vanishing Weyl tensor and positive scalar curvature, so

$$\frac{1}{2} \frac{d}{dt} \|F^+\|_{L^2}^2 = -\|\nabla F^+\|_{L^2}^2 + \langle \{F^+, F^+\}, F^+ \rangle - 4 \|F^+\|_{L^2}^2.$$

The only potentially dangerous term is the cubic one. It is estimated by

$$|\langle \{F^+, F^+\}, F^+ \rangle| \leq C \|F^+\|_{L^4}^2 \|F^+\|_{L^2},$$

while Sobolev inequalities give

$$\|F^+\|_{L^4}^2 \leq C(\|\nabla F^+\|_{L^2}^2 + \|F^+\|_{L^2}^2).$$

Since $\|F^+(t)\|_{L^2} \rightarrow 0$, the cubic term is eventually lower order and can be absorbed into the negative terms on the right-hand side. One then obtains

$$\frac{d}{dt} \|F^+\|_{L^2}^2 \leq -6 \|F^+\|_{L^2}^2$$

for all sufficiently large t , and hence

$$\|F^+(t)\|_{L^2} \leq Ce^{-3t}$$

for large t .

Finally, the exponential decay of $\|F_A^+(t)\|_{L^2(S^4)}$ can be upgraded in two steps. First, applying the local split ε -regularity theorem on unit backward parabolic cylinders for $t \geq 1$, one obtains

$$\|F_A^+(t)\|_{L^\infty(S^4)} \leq Ce^{-3t}.$$

Since $\|F_A^+(t)\|_{L^\infty}$ is therefore integrable in time, the continuation estimate (26) implies a uniform bound for the full curvature:

$$\sup_{t \geq 1} \|F_A(t)\|_{L^\infty(S^4)} < \infty. \quad (29)$$

With this global curvature bound, the Uhlenbeck convergence is in fact smooth convergence modulo gauge, and A'_∞ is a connection on the original bundle.

One of Alex' papers has the following estimate, for $t > t_1$, we have

$$\int_{t_1}^t \|D^*F(s)\| ds \leq Ce^{-3t_1}.$$

It then follows from (29) and a Lipschitz-type estimate of connection that as $t \rightarrow \infty$, $A(t)$ converges exponentially in H^k for all k to some smooth instanton A_∞ , it also follows that A_∞ is smooth gauge equivalent to A'_∞ .

Remark 4.1. *The main line of the argument is now clear. The gap condition is stated in terms of Yang–Mills connections, but the initial data are arbitrary. The local split ε -regularity theorem bridges this gap. It turns low L^2 self-dual energy into local pointwise control of F^+ . This prevents finite-time blow-up and gives a global flow. After that, compactness yields a Yang–Mills limit, and the definition of the gap forces this limit to be anti-self-dual. Finally, the geometry of the round sphere gives exponential convergence.*

In this way one obtains the following picture: every connection on S^4 whose energy lies below the first non-instanton Yang–Mills gap flows, under Yang–Mills flow, to an instanton.